

Sharpe Invariance of Static Hedge Ratios and Regime-Conditional Crude Hedging: Evidence from Indian Refiners

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Abstract

We document a Sharpe-invariance property of static hedging: hedge-relative savings scale linearly in the hedge ratio, so every constant-coverage policy delivers an identical savings-Sharpe — the choice among static hedges is one of scale, not risk-adjusted quality. Only state-dependent coverage can move that ratio. We exploit this in a 192-month backtest (2010–2025) of nine policies on Brent crude priced in Indian rupees, the exposure of Indian refiners whose disclosed coverage spans two orders of magnitude. A regime-aligned policy hedging more heavily in calm regimes raises the savings-Sharpe from 0.092 to 0.114 under a look-ahead-free rolling-quantile classifier, at shallower peak drawdown. A full-sample Markov classifier amplifies the edge to 0.182, but strict walk-forward re-estimation eliminates it: real-time crisis labels lag the crashes they detect. Bootstrap intervals straddle zero, and the edge inverted after 2020. We recommend conservatively sized regime-conditional hedging where a classifier can be operated.

Keywords: crude oil hedging; regime-switching; Indian refiners; Sharpe invariance; volatility regimes; Brent-INR.

JEL classification: Q40, G11, G32, C58.

1. Introduction

India is the world's third-largest crude oil consumer, importing approximately 4.6 million barrels per day in 2024–25 — roughly 88% of total petroleum demand. The country's four largest refiners — Indian Oil Corporation, Bharat Petroleum, Hindustan Petroleum, and Reliance Industries — collectively process more than four million barrels per day, generating annual P&L sensitivities of tens of billions of rupees to crude-price movements. Despite this exposure, disclosed commodity hedging coverage across the four firms varies by a factor of more than two hundred: from 0.07% (Indian Oil Corporation) to 16.50% (Hindustan Petroleum) of physical exposure, with no firm publicly articulating a rule connecting hedge intensity to observable market conditions. The variance reflects institutional preference, not analytical policy.

This paper asks a structural question motivated by that variance: what hedging policy should a major Indian refiner follow, and on what evidence? The question divides naturally into two parts. First, can directional forecasting of Brent-INR — the natural exposure series, defined as the

international Brent crude benchmark valued at the prevailing INR/USD exchange rate — produce reliable enough signal at policy-relevant horizons to drive a forecast-based hedging policy? Second, conditional on the answer to the first question, what class of rule-based hedging policy maximises risk-adjusted performance over an empirically meaningful backtest window?

Our answer to the first question is negative in a precise sense. Across a forecast horse-race spanning seven model specifications — ARIMA, SARIMA, Prophet (vanilla and with macroeconomic regressors), XGBoost, naive random-walk, and seasonal naive — fit on the 1987–2026 Brent-INR series and evaluated walk-forward over 257 monthly forecasts beginning in 2005, no specification produces statistically significant improvement over the naive baseline at conventional 5% significance thresholds. The result holds at 1, 3, 6, and 12-month horizons. At the 1-month horizon, two machine-learning specifications (XGBoost and Prophet with macroeconomic regressors) are significantly worse than naive under Diebold-Mariano tests.

The negative result on forecasting redirects the policy question toward rule-based hedging, where coverage is conditioned on observable market state rather than on a directional view. Within this class, we establish a mathematical result that constrains what is even possible: any static hedging policy — a constant hedge ratio applied uniformly across the sample — produces an identical Sharpe ratio on hedge-relative cumulative savings, regardless of the coverage level chosen. The result follows from positive-scalar scaling of returns: if R denotes monthly hedge-relative savings and h a static hedge ratio, then $\text{Sharpe}(hR) = h \cdot E[R] / (h \cdot \sigma(R)) = \text{Sharpe}(R)$. All four disclosed refiner-style policies and the three benchmark Static-30/50/70% policies we construct produce an identical annualised Sharpe of +0.092 to six decimal places over the 2010–2025 backtest.

We construct two independent regime classifiers — a rolling-quantile classifier on trailing six-month realised volatility, and a Markov-switching specification with two latent states — and backtest nine candidate hedging policies on 192 monthly observations from January 2010 to December 2025. A heavy-in-calm, light-in-crisis structure (70% coverage in calm regimes, 30% in high-volatility regimes) generates an annualised savings-Sharpe of +0.114 under the look-ahead-free rolling-quantile classifier and +0.182 under Markov-switching, against the static +0.092 baseline. Block-bootstrap confidence intervals on the inter-policy spread straddle zero, and the regime-conditional edge inverts post-2020 when high-volatility months coincide with rising rather than falling drift — the Russia-Ukraine supply-shock period.

2. Literature

Commodity-price forecasting. The forecast-evaluation methodology builds on the canonical finding that oil-price forecasting at monthly and longer horizons is unusually difficult relative to other macro-financial series. Alquist, Kilian, and Vigfusson (2013), surveying the field for the *Handbook of Economic Forecasting*, document that no-change forecasts of Brent and WTI are difficult to beat across a wide range of structural, time-series, and machine-learning specifications. Our contribution is the first comprehensive forecast horse-race on the Brent-INR series specifically — the international Brent price valued in domestic rupees, which is the relevant exposure for Indian refiners but distinct from both Brent-USD and the Indian Basket.

Regime-switching classification. The Markov-switching specification builds on Hamilton (1989), with the variance-switching extension following Hamilton and Susmel (1994). Our two-state Markov fit on monthly Brent-INR log returns isolates an asymmetric structure — a 92.5%-prevalent calm regime with positive drift (+1.21% per month) and a 7.5% rare crisis regime with sharply negative drift (−2.87% per month).

Hedging theory and the savings-Sharpe metric. The hedging-theoretic contribution is the explicit articulation of a Sharpe-invariance property. The classical minimum-variance-hedge-ratio framework — Johnson (1960) and Ederington (1979) — derives the variance-minimising static hedge ratio for the combined physical-plus-financial portfolio. By contrast, the savings-Sharpe metric we employ is mathematically invariant under positive-scalar scaling of the hedge ratio. The invariance result in the savings-Sharpe metric is, to our knowledge, novel. The closest precedent is Alizadeh, Nomikos, and Pouliasis (2008), who develop a Markov regime-switching approach for hedging energy commodities.

3. Industry context

India's four largest refiners, facing structurally identical crude-price exposure, hedge that exposure in proportions that differ by more than two orders of magnitude (Table 1). The figures are taken from the FY 2024-25 Integrated Annual Reports of Reliance Industries (RIL), Indian Oil Corporation (IOC), Bharat Petroleum (BPCL), and Hindustan Petroleum (HPCL). HPCL hedges roughly 16.5% of its throughput and RIL roughly 11%, while BPCL hedges about 2.3% and IOC

— the largest refiner by throughput — about 0.07%. Across approximately 1,500 pages of disclosure, no refiner specifies a rule connecting hedge intensity to observable market conditions.

Refiner	Disclosed coverage (%)	Coverage type
Indian Oil Corporation (IOC)	0.07	Operational (refined product / shipping)
Bharat Petroleum (BPCL)	2.30	Operational only
Reliance Industries (RIL)	11.00	Mixed (crude price + product)
Hindustan Petroleum (HPCL)	16.50	Mixed (crude price + product)
Sector average (volume-weighted)	≈ 7.0	—
Spread (max / min)	200×	—

Table 1. Disclosed commodity hedge coverage, four largest Indian refiners (FY 2024-25). Source: Integrated Annual Reports FY 2024-25. None of the four refiners discloses a rule connecting hedge intensity to observable market conditions.

4. Data

4.1 Sources and the Brent-INR construction

The analysis uses three public daily series from the Federal Reserve Economic Data (FRED) repository: the Europe Brent spot price in US dollars per barrel (DCOILBRENTU), the West Texas Intermediate spot price (DCOILWTICO), and the Indian-rupee-to-US-dollar exchange rate (DEXINUS). Aligned on common trading days, the daily sample runs from 20 May 1987 to 11 May 2026 (10,169 observations). The variable of interest is Brent priced in Indian rupees:

$$Brent-INR(t) = Brent-USD(t) \times (INR/USD)(t).$$

This is the rupee outlay per barrel faced by an Indian refiner. It carries two sources of risk — the dollar oil price and the exchange rate — and over the monthly sample correlates with dollar Brent at 0.92.

4.2 Frequency and the backtest window

The daily series are downsampled to monthly frequency by averaging within each calendar month. This yields 469 monthly observations from May 1987 to May 2026. The hedging backtest runs over January 2010 to December 2025 — 192 months.

4.3 Return and volatility structure

Over 39 years Brent-INR rose more than forty-fold, from ₹236 to ₹10,371 per barrel (sample mean ₹2,924). Returns are mildly autocorrelated at lag one (+0.26, Ljung-Box $p < 0.001$). Volatility clusters strongly, with extended calm periods punctuated by episodes of sharply elevated variance — the foundation of the regime analysis.

Series	Mean	Std. Dev.	Min	Max
Brent (USD/bbl)	51.28	32.83	9.83	133.21
WTI (USD/bbl)	49.58	29.34	11.33	133.89
INR per USD	49.44	19.94	12.72	94.74
Brent-INR (INR/bbl)	2,923.70	2,394.41	182.56	11,051.45
Brent-INR 1-mo log return (%)	0.81	9.77	-52.09	45.50

Table 2. Summary statistics, monthly series, May 1987 – May 2026 (n = 469).

5. Forecast-based hedging is ruled out

A companion study evaluates seven forecasting models on the monthly Brent-INR series: a random-walk (naive) baseline and seasonal-naive baseline; ARIMA(1,1,1) and seasonal ARIMA; Prophet with and without exogenous regressors; and XGBoost trained on twenty-three engineered features. Each is evaluated by walk-forward backtesting at horizons of one, three, six, and twelve months, assessed by the Diebold-Mariano (1995) test with Harvey-Leybourne-Newbold (1997) small-sample correction.

The result is uniformly negative. At the one-month horizon the most accurate model, ARIMA(1,1,1), achieves a MAPE of 6.98% against the naive baseline's 7.29% — an edge of 0.31 percentage points indistinguishable from zero ($p = 0.28$). XGBoost and Prophet with regressors are significantly worse than naive ($p = 0.003$ and $p = 0.006$). At horizons of three months and beyond, none of the thirty pairwise comparisons is significant at the 5% level.

Comparison vs naive	DM statistic	p-value	Verdict
ARIMA(1,1,1)	-1.08	0.281	Tie
SARIMA(1,1,1)(1,1,1,12)	-0.82	0.411	Tie
XGBoost	+2.77	0.006	Naive wins
Prophet + regressors	+2.96	0.003	Naive wins

At 3-month horizon (10 pairs)	—	all $p \geq 0.10$	All tied
At 6-month horizon (10 pairs)	—	all $p \geq 0.10$	All tied
At 12-month horizon (10 pairs)	—	all $p \geq 0.10$	All tied

Table 3. Diebold-Mariano pairwise tests vs naive baseline, 1-month horizon.

Rank	Feature	Gain importance
1	Brent-INR, lag 1	0.520
2	Brent-INR, lag 3	0.197
3	Brent-USD, lag 1	0.085
4	Year	0.080
5	Brent-INR, lag 6	0.042
6	Brent-INR, lag 12	0.009
	Sum of top 6 (of 23 features)	0.933
	Concentration on lags 1 + 3	0.717

Table 4. XGBoost feature importance. Of 23 engineered features, 71.7% of importance loads on Brent-INR's own first and third lags.

6. Regime classification

6.1 The rolling-quantile classifier (primary)

For each month the trailing six-month sample standard deviation of Brent-INR monthly log returns is computed. A month is labelled 'high-volatility' if its trailing volatility equals or exceeds the full-sample median $\sigma^* = 0.0687$ (23.8% annualised), and 'calm' otherwise. For the backtested policy an expanding-window variant is used, excluding all look-ahead. Over January 2010 – December 2025 the classifier produces a 114/78 split: 114 calm months and 78 high-volatility months. The policy-relevant drift following each label: +₹15.1/bbl after calm months against +₹5.9/bbl after high-volatility months.

6.2 The Markov-switching classifier (robustness)

A two-state Markov-switching model (Hamilton, 1989) is fit to the monthly log returns with switching mean and variance, estimated by the EM algorithm. The calm regime covers 92.5% of

months, with mean monthly return +1.21% and annualised volatility 24.7%; the crisis regime covers 7.5% of months with mean return −2.87% and annualised volatility 76.3%.

6.3 Classifier agreement

Cohen's κ between the rolling-quantile and Markov labels is 0.14 — only slight agreement on the Landis-Koch (1977) scale (raw agreement 57%). The robustness claim worth drawing is that the same directional policy conclusion survives both definitions.

6.4 A three-state extension (diagnostic)

The three-state Markov fit decomposes the crisis regime into a rare crash state (mean −11.24% monthly) and a high-variance supply-shock state with positive drift (mean +1.93%, annualised volatility 91.8%). This decomposition explains the post-2020 policy inversion documented in Section 8.4.

Spec.	Regime	Mean (%/mo)	Ann. std (%)	Duration (mo)	Share (%)
K=2	Low (calm)	+1.21	24.7	27.1	92.5
K=2	High (crisis)	−2.87	76.3	3.0	7.5
K=3	Low (crash)	−11.24	21.3	1.6	7.7
K=3	Mid (normal)	+2.07	22.5	12.9	88.0
K=3	High (supply-shock)	+1.93	91.8	2.4	4.3

Table 5. Markov-switching regime statistics, $K = 2$ and $K = 3$. $K=2$ adopted as primary on BIC grounds.

7. Policy framework and backtest design

7.1 The refiner's problem and sign convention

The hedger is a crude consumer, not a producer. At the end of month $t-1$ the refiner locks in a fraction h of the coming month's barrels at the prevailing spot price $P(t-1)$; the per-barrel hedge result is $h \times [P(t) - P(t-1)]$. No future information enters the hedge decision; the backtest is free of look-ahead by construction.

7.2 The nine candidate policies

Nine policies are evaluated: four static at disclosed coverage levels (0%, 2.3%, 11%, 16.5%), three round-number static benchmarks (30%, 50%, 70%), and two regime-conditional policies: P8 (70% calm / 30% high-volatility) and its mirror P9 (30% calm / 70% high-volatility).

7.3 The savings-Sharpe invariance result

For any constant hedge ratio h , the savings series is h times the unhedged return series, so:

$$\text{Sharpe}(h \cdot R) = h \cdot E[R] / (|h| \cdot \sigma(R)) = E[R] / \sigma(R) = \text{Sharpe}(R).$$

Every static policy, at any coverage level, therefore produces an identical savings-Sharpe. Improving it requires a policy whose ratio varies with market state — the regime-conditional class.

8. Results

8.1 Static hedging cannot change the savings-Sharpe

Every static policy in the backtest produces an identical annualised savings-Sharpe of 0.092, agreeing to six decimal places. Cumulative savings rise monotonically with coverage, from ₹50/bbl at 2.3% to ₹1,521/bbl at 70%, but the ratio of return to risk is invariant.

8.2 Regime-conditional hedging departs from the invariant ray

Under the primary rolling-quantile classifier, the regime-aligned policy P8 (70% calm, 30% high-volatility) achieves an annualised savings-Sharpe of 0.114, against the static value of 0.092 — an edge of +0.022. Its mirror P9 falls to 0.061. Under full-sample Markov labels P8 rises to 0.182; however, this amplification is eliminated by strict walk-forward re-estimation (Section 8.6).

Policy	Cum. ₹/bbl (RQ)	Sharpe (RQ)	Cum. ₹/bbl (Markov)	Sharpe (Markov)
BPCL-style (2.3%)	50	+0.092	50	+0.092
RIL-style (11%)	239	+0.092	239	+0.092
HPCL-style (16.5%)	359	+0.092	359	+0.092
Static 30%	652	+0.092	652	+0.092
Static 50%	1,087	+0.092	1,087	+0.092
Static 70%	1,521	+0.092	1,521	+0.092
P8 (70% calm / 30% crisis)	1,339	+0.114	2,702	+0.182

P9 (30% calm / 70% crisis)	835	+0.061	−528	−0.053
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Table 6. Hedging policy backtest results, January 2010 – December 2025 (n = 192). RQ = rolling-quantile K=2 (expanding-window, look-ahead-free); Markov = Markov-switching K=2 (full-sample labels).

8.3 The magnitude of the advantage is not statistically robust

A stationary block bootstrap (5,000 replications, 6-month mean blocks) produces confidence intervals that straddle zero for every inter-policy spread. P8's cumulative savings carries a 95% CI of [−₹4,982, +₹7,325] with only a 67% bootstrap probability of being positive.

Comparison	Point (₹/bbl)	95% CI (₹/bbl)	P(spread > 0)
P8 vs Static 50%	+252	[−1,775, +2,296]	0.60
P8 vs Static 70%	−183	[−3,540, +2,808]	0.48
P8 vs P9 (mirror)	+504	[−3,550, +4,593]	0.60
P8 vs HPCL (16.5%)	+980	[−3,602, +5,233]	0.67

Table 7. Bootstrap confidence intervals on cumulative-savings spreads. All four CIs straddle zero.

8.4 The advantage inverted after 2020

Splitting the backtest at March 2020: pre-COVID, P8 Sharpe 0.125 vs static 0.044 (edge +0.081); post-COVID, P8 Sharpe 0.098 vs static 0.149 (edge −0.051). The 2022 Russia-Ukraine supply shock produced high-volatility months with strongly rising prices, under-hedging P8 precisely when coverage was most needed.

8.5 Drawdown and cost

P8 delivers 88% of Static 70%'s cumulative savings (₹1,339 vs ₹1,521) while incurring only 70% of its peak drawdown (−₹2,807 vs −₹4,003/bbl). Break-even transaction cost is approximately 25 basis points per month of notional against the unhedged baseline.

8.6 A strict walk-forward Markov test

Under strict walk-forward re-estimation (model re-fit at each month t on returns through t−1 only), the Markov classifier flags 33 crisis months vs the smoothed specification's 13, arriving late. Walk-forward P8 earns a Sharpe of 0.067 — below the static 0.092. The full-sample Markov amplification is a look-ahead artifact. The only regime-conditional edge that survives genuine real-time classification is the rolling-quantile result (0.114).

9. Discussion and policy implications

9.1 What the evidence supports

Three findings are robust: (1) static hedging at any level delivers the same savings-Sharpe; (2) a regime-aligned policy improves the savings-Sharpe under both classifiers, explained by the calm-month drift premium; (3) the regime-aligned policy holds a shallower peak drawdown than static policies of comparable coverage. Two findings counsel restraint: the magnitude is not statistically distinguishable from zero (§8.3), and the edge inverted after 2020 (§8.4).

9.2 A conditional recommendation

Where a classifier can be operated: a regime-aligned structure sized below its in-sample optimum — heavier coverage in calm regimes (around 50%) and lighter in high-volatility regimes (around 15%) — using the expanding-window rolling-quantile rule. Where a classifier cannot be operated: a static policy at moderate coverage (30–50%), which sacrifices nothing in savings-Sharpe relative to any other static level.

9.3 The liquidity precondition

Hedge-book drawdowns translate directly into variation-margin calls. The regime-aligned policy's peak drawdown of $-\text{₹}2,807/\text{bbl}$ implies, for an IOC-scale refiner, a variation-margin exposure on the order of $\text{₹}5$ billion. A dedicated liquidity facility is a precondition, not a refinement. Replicating Brent-INR exposure requires two legs — a dollar-denominated crude hedge and a rupee-dollar currency overlay — each carrying its own basis risk and transaction cost.

10. Limitations

1. Sample-path dependency. The 192-month backtest reflects one realisation of Brent-INR. The post-2020 inversion is evidence that the 'high volatility implies negative drift' pattern is not structural.

2. Stylised hedge instrument. The backtest assumes frictionless monthly forwards. The Brent-INR exposure requires two legs in practice, each with its own basis risk and transaction cost.

3. In-sample classifier design. The classifier parameters (two states, median split, six-month window) were informed by full-sample inspection.

4. Brent as proxy for the Indian Basket. The public Indian Basket series ends in 2012, before the post-2020 inversion period. Post-2022 Russian Urals discounts may have widened the basis.

5. Volatility-only regime definition. A production classifier should incorporate fundamental signals — OPEC+ supply decisions, geopolitical-risk indices — to address the post-2020 failure mode.

6. Single-refiner abstraction. Optimal coverage may vary across IOC, BPCL, HPCL, and RIL given differences in refinery complexity and downstream margin structure.

11. Conclusion

This paper asked what rule should govern crude hedging for an Indian refiner, given that the rupee cost of crude cannot be forecast. The firm structural finding is that static hedging leaves the savings-Sharpe unchanged at any coverage level. Improving risk-adjusted hedging outcomes requires regime-conditional coverage — heavier in calm regimes, lighter in turbulent ones. The only classifier under which this edge survives genuine real-time operation is the expanding-window rolling-quantile rule. The magnitude of the edge is not statistically robust and inverted after 2020. The honest recommendation: adopt the regime-aligned structure where a classifier can be operated, sized conservatively, alongside a liquidity facility; hedge statically at moderate coverage otherwise. For a sector hedging a low single-digit fraction of its exposure, the structural case for hedging more, and more intelligently, is the durable finding.

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